

**Common Fibers Boat Project:
Accelerated UV Testing and Project-Specific Heated Tooling**

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1. Introduction

Composite materials have grown at a rapid rate across all types of industries including automotive, aerospace, and nautical. While the term “composite materials” spans a vast array of material, polymer matrix composites such as CFRP or FGRP pose advantageous mechanical characteristics as well as specific-strength. Unfortunately, these composites come at a hefty cost when compared to that of aluminum or steel. In the nautical world, almost all speed boats or pleasure craft are constructed in large part or entirely out of composite materials, whether that be carbon or fiber glass. Due to the excessively high price that modern-day boats cost, with top end tow boats like the Malibu Wakesetter series coming in as high as roughly \$450,000, there leaves a market of individuals interested in boating, but lacking the means to purchase such expensive vehicles. Thus, a multi-purpose boat poses considerable attraction, as it can fit the needs of many people without breaking the bank. To design and construct this style boat at a reasonable price, material selection and manufacturing processes need to be streamlined to minimize maintenance costs as well as labor costs. The objective of the Common Fibers boat project to design and construct a multi-purpose boat capable of a variety of water activities, without requiring excess spending from interested parties. To achieve this major objective, smaller objectives needed to be met, including understanding the impact of long-term UV exposure on various candidate materials, as well as implementation of heated tooling on a boat gunnel tool.

2. Background & Objectives

The Boston Whaler hull design and outboard motor application offered a good starting point for the Common Fibers design. The shape of the hull can maintain a level deck in choppy water - ideal for fishing, sleeping, or general stationary water time, as well as take on higher speeds without losing control of steering.

2.1 Tooling Expansion

Reducing cure time for composite parts saves time and money. When designing tools that are expected to undergo heating, it is important to consider that all materials behave uniquely when exposed to temperature changes. While Common Fibers builds many of its molds out of fiberglass, that introduces expansion challenges when using a heated fiberglass mold on a carbon fiber part. To better design molds, research is required to understand material coefficients of thermal expansion (CTEs).

2.2 Accelerated UV Testing Objective

Reducing maintenance costs includes understanding the impact of the environment the composites will be operating in. The objective of accelerated UV testing was to understand the effect of long-term UV exposure on the cosmetic and mechanical properties of the different raw and coated composites that Common Fibers uses in its constructions.

2.2 Boat Gunnel Prototype Objective

To increase production capability, testing on prototype tooling was required to understand optimal design parameters. The objectives for the prototype boat gunnel include testing the boat

rail system that will be used to incorporate add-on features, and testing resistive heated tooling design.

3. Tooling Expansion

3.1 Research

To incorporate a heating system into the tooling in Section 4, an understanding how different materials change under temperature is important. In an ideal situation, the same material would be used for the mold and the part as it would result in the same thermal expansion rate between both components. This uniformity would significantly reduce the risk of warping or other defective manufacturing results. The challenge that making carbon fiber parts poses is the cost to create a mold out of carbon fiber is much higher than to create one with fiberglass.

Carbon fiber can be manufactured in a variety of ways which makes it difficult to nail down one number for its coefficient of thermal expansion (CTE). Twill designed plies have quasi-isotropic CTE in the axial and transverse directions of roughly $1.0\text{--}2.5 \times 10^{-6}/^{\circ}\text{C}$. This CTE is much lower than aluminum or steel which are about $23 \times 10^{-6}/^{\circ}\text{C}$ and $12 \times 10^{-6}/^{\circ}\text{C}$, respectively. It's important to note that in unidirectional carbon fiber plies, the axial CTE can actually be negative. Fiberglass has a CTE of about $5 \times 10^{-6}/^{\circ}\text{C}$ and is considered generally isotropic. The volume (3/4oz vs 1 1/2oz) of fiberglass ply does not substantially alter its CTE by itself, but the change in resin required to saturate the layer does. The vinyl ester resin used to saturate the fiberglass used for the boat gunnel mold discussed in Section 5 has a large CTE range of 50 to $120 \times 10^{-6}/^{\circ}\text{C}$. Thus, when creating fiberglass tools using this resin, it's preferable to minimize resin use to reduce the impact of its high CTE because the fiberglass counterpart is much lower.

The EG 160 Epoxy Tooling Gelcoat, EMP160 Epoxy Paste, and EL160 Epoxy Resin manufactured by Easy Composites do not have public publishings for their CTEs. However, estimates put the EG160 in the $45\text{--}60 \times 10^{-6}/^{\circ}\text{C}$ range, the EMP160 in the $25\text{--}35 \times 10^{-6}/^{\circ}\text{C}$ range, and the EL160 in the $50\text{--}65 \times 10^{-6}/^{\circ}\text{C}$ range. It is stated that, when used in the correct proportion with carbon fiber, the EL 160 can significantly reduce in expansion as the carbon fiber dominates axial expansion.

4. Accelerated UV Testing

4.1 Methods & Materials

A variety of materials were used to create panels that represented the different combinations of materials used at Common Fibers. Rovicore, dry fiber, and prepreg panels were created. Rovicore panels consisted of 1 layer of Rovicore 600, and were infused with plain vinyl ester, vinyl ester with a gelcoat dye colorant, and Proxima epoxies. The dry fiber panels consisted of 6k, G-flow, and 12k plies. They were infused with CPD epoxy. Another set of dry fiber panels, created at the Redmond location, consisted of 6k, G-flow, and 12k and were infused with either a vinyl ester or a polyester. The prepreg panels consisted of 3k and 2 12k plies. The carbon fiber panels were sprayed with various in-mold coatings (IMCs) and finish sprays - see Tables 1 and 2 for further detail. Images of the control samples for each panel are found in Appendix A, below.

Each vacuum area had to pass a 15 minute pressure test prior to infusions.

All samples were done in their own vacuum areas to reduce risk of contamination. Each type of sample required 4 6in x 6in squares, so each vacuum area was roughly 18in². Figure 1 shows a typical infusion vacuum set-up. To maximize UV testing accuracy, overlaying carbon fiber plies was avoided when possible - some minor overlay exists in the prepreg panels, but was only done to the extent recommended, which was at least 1in of overlap.

4.1.a. Proxima Resin System

To prepare Proxima resin, select a desired amount (specific test amounts for these panels are found in Table 1) and add 2% weight of its catalyst - make sure to thoroughly shake the catalyst before adding. Mix thoroughly and place in a vacuum tank to degas for at least 30 minutes. Slowly release pressure and carefully move the Proxima into position for infusion. Take care to not shake the container before infusion, as it is vital to keep the liquid degassed of oxygen. Once infused, the part went into the oven at 190F for 40 minutes to cure.

4.1.b. Vinyl Ester Resin System

The vinyl ester resin was mixed with 2% by volume of CHP. It is important to note that vinyl ester resin used with fiberglass for mold-making is mixed with MEKP, but those should not be mixed up. After a thorough mix, it is ready for infusion. It tended to flow quickly, as evident in infusion times shown in Table 1, so minor flow control on the inlet tube was useful.

4.1.c. CPD Epoxy Resin System

To infuse parts with CPD epoxy, the resin to catalyst ratio posted in the shop was used. Shown in Table 2, resin amounts varied from 300g to 350g and the catalyst varied accordingly. The catalyst is roughly 3%-3.5% by weight of the CPD epoxy. Ideal infusion times are closer to 12-15 minutes. Once the epoxy is mixed, it needs to be infused, as it has a short cup life.

4.1.d. Prepreg System

While prepreg did not involve resin infusion, once laid up, release film and breather were put over top before the part underwent a 24 hour debulking period. Following debulking, the part was cured at 200F for 2 hours.

4.2 Manufacturing Details & Panel Results

Proper fiber saturation results in more accurate test results. To achieve this, data was kept on infusions and flow was adjusted manually to allow for better infusion processes. At the time of lay-ups - mid-July 2026, ambient room temperature was high, so infusion times had to be balanced with epoxy cure time, particularly in regards to the CPD epoxy.

	Proxima - R3	Proxima w Dye - R2	Vinyl Ester - R1	Vinyl Ester w White Gelcoat Dye - R4
Infusion time	~8'	~6'	~8'	~10'
Resin amt	250g	350g	400mL	425mL
Hardener amt	5g	7g	8mL	8.5mL

Epoxy Adds.	No	TiO2 powder	No	Gelcoat Dye - white
IMC type	N/A	N/A	N/A	N/A
Finish spray	N/A	N/A	N/A	N/A

Table 1. Rovicore 600 infusions. Proxima with Dye (R2) was a failed infusion, as it did not show noticeable change in color from the regular Proxima infusion.

	CPD epoxy - raw - I1	Berg IMC over CPD epoxy - I2	PPG over Berg IMC over CPD epoxy - I3	PPG IMC over CPD epoxy - I4
Infusion time	~8'	12' 34"	15' 32"	10' 32"
Resin amt	300	350	350	350
Hardener amt	95	105	105	105
IMC type	N/A	Bergolin	Bergolin	PPG Omni Plus
Finish spray	N/A	N/A	PPG Omni	N/A

	PPG over PPG IMC over CPD epoxy - I5	Imedge Gelcoat over CPD Epoxy - I6	SunShield over CPD Epoxy - I7
Infusion time	9' 17'		17' 30"
Resin amt	350	300	350
Hardener amt	105	95	105
IMC type	PPG Omni Plus	Imedge 120x Clear Gelcoat	N/A
Finish spray	PPG	N/A	Sunshield

Table 2. 6k/G-Flow/12k dry fiber and CPD epoxy infusions.

Proxima is a unique polyolefin resin in that it is a thermoset, versus the typical thermoplastic. Thus, it's a fundamentally different system due to polymer cross-linking and its inability to be melted and reformed. The low viscosity of the Proxima resin resulted in very fast infusion times. While it is ideal to extend the infusion time closer to at least 10 minutes, it was important to reduce backflow of the resin as it moved through the tubing, so the 8 minutes noted in Table 1 satisfied the saturation of the material. The resulting finish, seen in Appendix A.1, was a translucent beige. From a cosmetic boat hull finish standpoint, it would be ideal to see a more prominent color.

Dying the Proxima resin proved to be a harder task than its vinyl ester counterpart because of its thermoset properties. Carbon dust and Titanium Oxide (TiO2) powder were both experimented with, directly mixed with the resin system prior to degassing. Neither resulted in promising results as the powder was heavier than could be suspended within the resin. Carrier

fluids were researched as a means of “pre-mixing” the powder before adding it to the Proxxima. The most promising result was the high-purity, synthetic isoparaffinic hydrocarbon chemical called Isopar. Because it is a hydrocarbon base - like Proxxima - it has the best chance of mixing with both the powder and resin without poisoning the catalyst.

The vinyl ester and CPD epoxy exhibiting no unusual infusion complications. The vinyl ester also mixed well with the gelcoat dye used and produced a creamy white finish, which could be of cosmetic interest for industry application.

4.3 Sub-project Next Steps

These panels will be sent for accelerated UV testing via facilities from Polytek Development Co. Each panel type was cut into 4 squares, thus, one square will be kept as a control to compare any UV-induced changes against.

Following UV testing, the panels are expected to be sent to Cal Poly for mechanical testing. Test plans are not confirmed but due to the dimensions of the panels, flexural testing is the most promising avenue given the equipment they have available. While flexural testing will provide useful data such as the relaxation modulus and its change over time with respect to different composites, tensile testing will provide better data on Young’s Modulus and Poisson's ratio. If Cal Poly is not able to conduct tensile testing, one local option to outsource that work is through [Element](#). The company has test facilities in Bothell, and they do mechanical testing of all types of materials.

5. Boat Gunnel Prototype Objective

Ahead of full-scale boat tool manufacturing, a prototype boat gunnel plug was made out of 20lb tooling board foam. This gunnel will be used to test add-on designs and the effectiveness of the rail system that is to be incorporated throughout the whole boat.

5.1 Methods & Materials

5.1.a. Gunnel Plug

The plug was cut from 20lb tooling foam. Adhering the foam to the skid used vinyl ester and Cab-O-Sil in a 1:1 ratio. To calculate resin need, calculate the surface area of the board and multiply by .75mm. Then convert to milliliters. Clamp the foam to the skid using weights and clamps until dry.

The primer was mixed with 10% primer volume MEKP catalyst and 10% primer volume Duratec Reducer. After the primer was dry, Bondo was used on divots or imperfections. Bondo rested for at least 10-15 minutes before sanding was done on it. Sanding was done at 80 grit, then 120 grit. After sanding, the plug was cleaned with acetone and Tacky Cloth before spraying a second round of primer. In the second primer coat, a clear coat additive was added for a slightly glossier finish. With the Duratec Primer, add 10% primer volume of the Duratec Reducer, 10% primer volume of Duratec Additive, and 2%-2.5% primer volume of MEKP catalyst. Spray a glossy finished, thick coat.

The second primer coat was sanded from 150 dry to 320 wet, then 400 wet. Following sanding, the tool was waxed with a TR High Temp Mold Release (and whipped off) 6 times, with 30 minutes between applications.

After the wax had set, the gelcoat was sprayed to a thickness of 30 (units). The gelcoat is mixed with 10% volume of acetone and 2% gelcoat volume of MEKP catalyst.

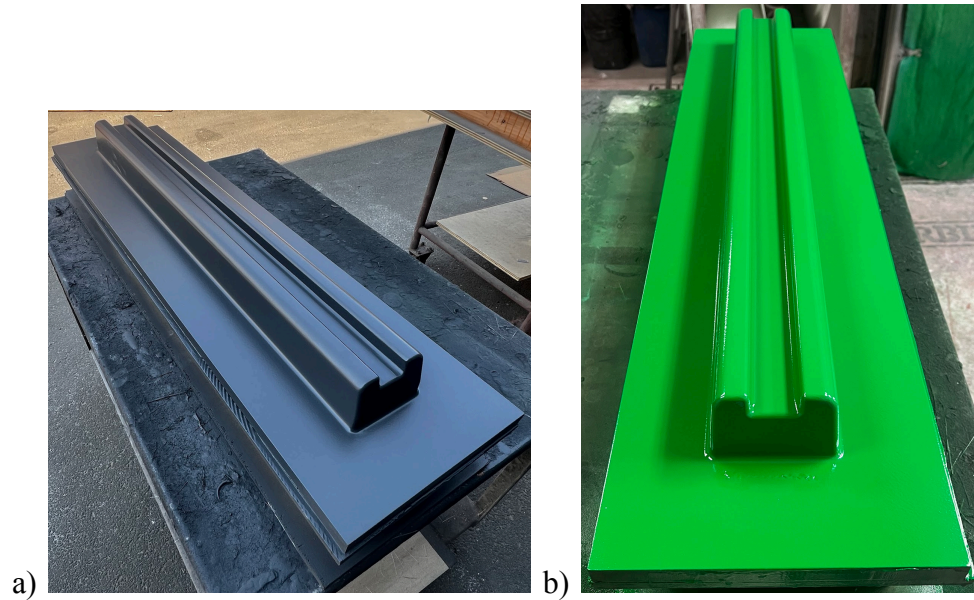


Figure 1. Boat Gunnel plug with a) 1st layer of Duratec Primer sprayed, and b) Duratec gelcoat applied.

5.1.b. Gunnel Tool

In an effort to manufacture boat components at a faster, and more cost effective, rate, the gunnel prototype tool contributes to early-stage testing for the boat's rail system as well as heated tooling design throughout more complex tool geometries.

Following the gelcoat coating, shown in Figure 1, above, a putty was used to fill in some of the tighter radii highlighted in Figure 2.



Figure 2. Radius application zones of tooling putty.

Seven layers of fiberglass were then laid in 3 increments, 2 sets of 2 layers and 1 of 3. Increment 1 included a $\frac{3}{4}$ oz fiberglass sheet which aided in smoothing out the tighter radii prior to the $1\frac{1}{2}$ oz fiberglass application. Minor sanding - or scuffing - was done between sets, when hardened, to remove stray fibers.

After layer seven, a resistively-heated wire mesh was laid across the tool. Design of the wire mesh was driven using Isaac's calculator and is shown in Figure 3. To achieve a balanced amperage to volt balance, 4 elements of wire (from a wires positive to negative ends) were cut at roughly 140 to 150in. The wire used is a 26 gauge Nichrome 20/80 which then provides about 10 Ohms of resistance per element. The resulting voltage and amperage requirements are 76.21 and 7.4, respectively. It is preferable to reduce amperage as it mitigates electrical risk to technicians using the mold.

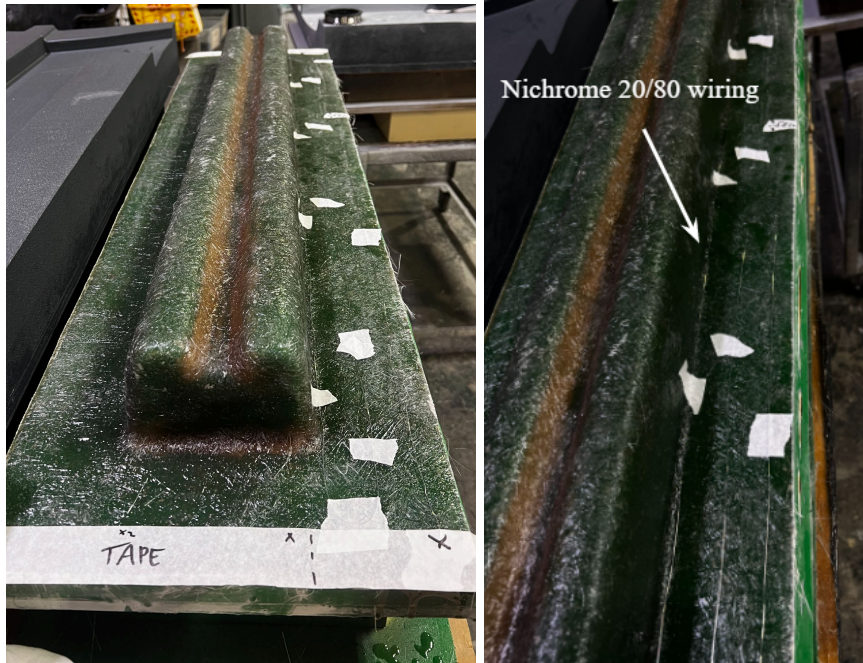


Figure 3. Representative wire layout for first take on heated gunnel tooling.

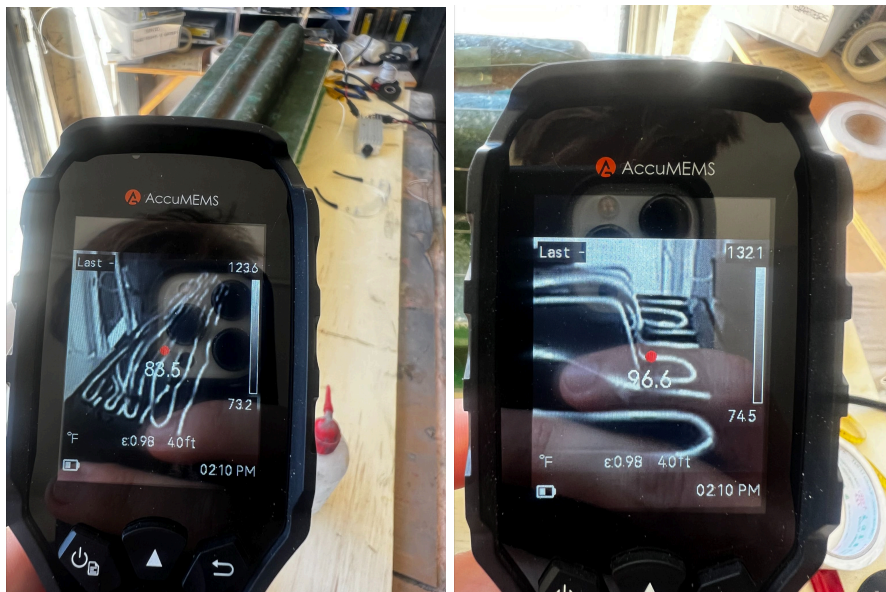


Figure 4. Open-to-air wire mesh showing successful heating of as high as 132.1F.

Prior to sealing the wiring into the tool, testing was done to confirm the parallel circuitry would work. Once the mesh is in place and tested, shown in Figure 4, 2 layers of 3/4oz fiberglass and 1 layer of 1 ½ oz are placed on top. It is recommended to tape off any electrical components that should not risk coming in contact with resin or fiberglass fibers. A mix of hot glue and high temperature tape was used to hold wiring in place during lay-up. While the tape did not adhere

well, taping over the length of the wire - shown in Figure 5 - resulted in much fewer voids and defects within the layers of fiber glass.



Figure 5. Red boxes highlight tape-covered wiring and red lines indicate non-covered wiring along the gunnel wall. The right image shows the higher number of air pockets and voids resulting from a lack of taping.



Figure 6. Heated element tested after fiberglass sealed it into the tool. Full heating capacity was not tested at this stage.

5.2 Lay-Up Results

5.2.a. Test 1

The first test was conducted with one layer of 600 Rovicore, infused with vinyl ester resin mixed with white gelcoat. I made a mistake in using MEKP as the catalyst when the vinyl ester required CHP, instead. Due to this, the product is not fully curing, even after the extensive heating the mold experienced, which is outlined below.

After lay-up and bagging was complete, the heating elements were turned on to a low voltage, around 85V, which seemed to be the lowest voltage usable to see heating. The heating was kept on during the pressure-leak test and turned off before infusion started. I used 1.5% resin volume of MEKP with the expectation that the higher mold temperature would induce a faster cure time. Due to the nature of the controller, the user can change input voltage as a means of affecting the heating intensity, and therefore, the temperature. After reaching high temperatures of 200F plus in some regions, see below in Figure 7, the voltage was turned to the minimum needed to maintain heating. Unfortunately, this still heated the elements faster than they dissipated heat, which caused it to overheat, smoke, and require shut down. The mold and part reached temperatures of around 300F, causing melting and mold damage, seen in Figure 8.



Figure 7. Temperature imaging via thermal gun during cure cycle after completion of infusion.



Figure 8. Bagging and part damage as a result of 300F hot spots. This image is on the gunnel wall corner and along the gunnel edge.

The high temperatures forced a shut down of the heating elements to maintain a safe work environment. Aside from using the correct catalyst, next steps for a second test would include keeping a closer eye on hot spot temperatures.

5.2.b. Test 2

Test 2 used 2% of the correct catalyst, CHP, with the same vinyl ester resin. White and red gelcoat dye was mixed into the resin prior to the addition of the catalyst. 1000mL of resin was used and 20mL of CHP was used (2% by volume).

During vacuum pressure testing, the voltage controller was kept much more conservative, keeping the mold around 105-120F. Infusion was conducted with the mold around 115-120F, and took just under 7.5 minutes. While this was faster than Test 1, the saturation was just as good. The progression of curing is shown in Figure 9. The controller was monitored carefully, and there was never one set voltage for the mold. Typically the controller kept the voltage between 70% to 85% of maximum (120V). Hardening occurred quickly, about 15 minutes into the cure, but slowed. Overall, the heating elements were on for roughly 1 hour. The part was then left to cool completely to avoid warping.

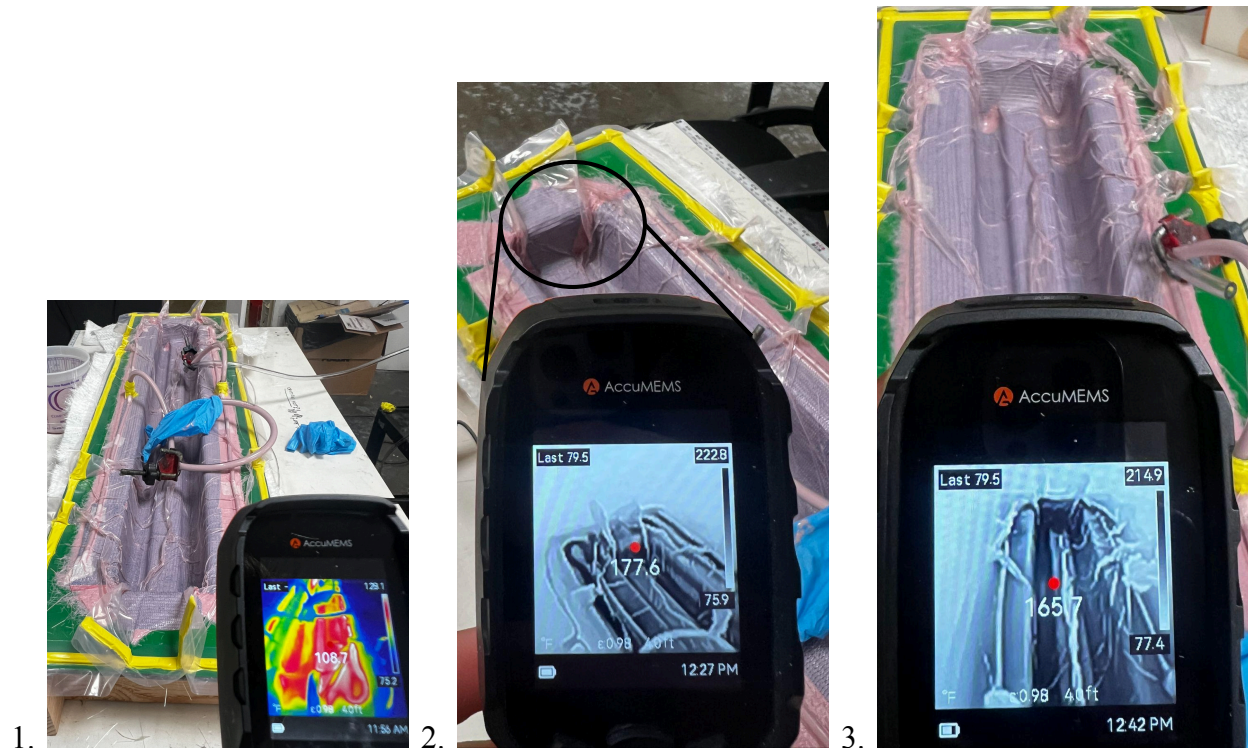


Figure 9. Thermal imaging as cure time progressed. Notably lower temperatures to start (image 1, around 125F), and higher temperatures near the end (image 3, closer to 220F).

Part extraction from the mold revealed more detail on the cure of the part. Cure in some parts of the gunnel were worse than others, insufficient areas are labeled in Figure 10. One contributor to poor cure in some areas could be due to mold damage from Test 1 (see Figure 11). The left side of the mold in the figure shows large air pockets which may have contributed to poor heating of the mold surface. There is also noticeable bridging along the rail edges. Better compaction of the Rovicore into radii is required.

In future infusions, a weighted roller could be used to push resin into the material for a better saturation. Additionally, shallower long ends may help with pulling the part out of the mold. Based on how the mold heated during testing, less putty use on the rail edges may be useful in keeping a more uniform temperature across all areas of the part.

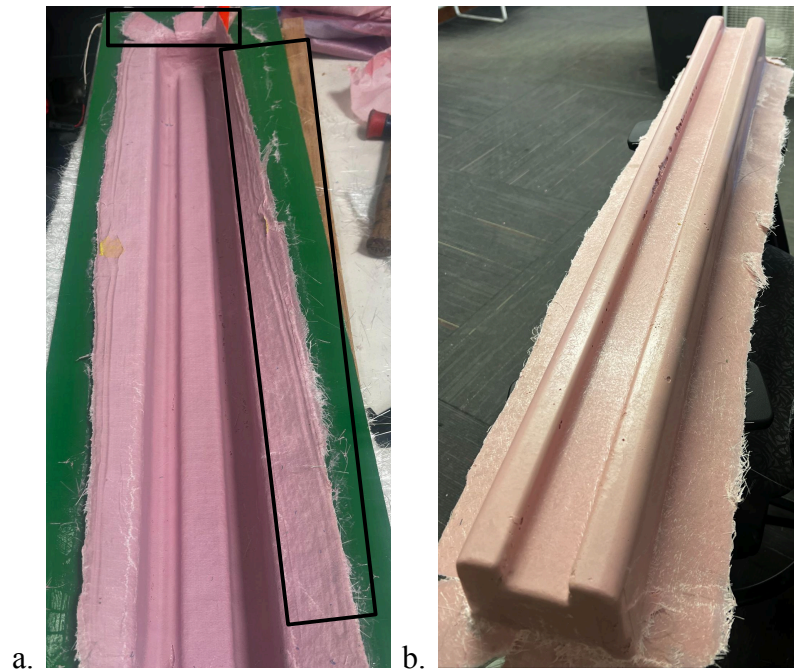


Figure 10. a) Insufficient cure on flanges resulted in flimsy material; b) finished part out of mold with notable bridging along rail edges.



Figure 11. Mold under-side post testing with severe damage due to overheating in Test 1.

5.3 Gunnel Prototype Next Steps

Refining the construction of heated tools is important in the efficiency of building boat components from time, financial, and space standpoints. The prototype gave a good comparative study between with and without high temperature tape. A better wire straightening tool may also assist in better wire lay-down to reduce air pockets along the heated elements. A very useful tool to incorporate into future heated tooling would be thermocouples. Experimenting with a thicker wire may also be an interesting experiment. The 26 gauge wires used in these tests run at a much higher temperature to heat the mold surface to the correct temperature. Using a thicker wire would lower the operating temperature of the wire itself without compromising the surface temperature. A risk of thicker wiring is the increase in amperage that comes with it, which could increase technician safety if not insulated in fiberglass properly.

Placing thermocouples within the mold near the tool surface would give a better idea of when to adjust voltage input. This incorporation would reduce risk of overheating like seen in Test 1. Further thermocouple and PID details are outlined in Section 6, below.

6. Conclusion & Next Steps

With the large-picture goal of manufacturing an affordable, versatile boat, the tooling expansion research, UV testing panels, and gunnel prototype are steps in the right direction. Utilizing what was learned from these tests (or what will be learned after UV testing is complete) will help increase efficiency in boat-making production as well as general shop production, and material selection.

The next steps for UV testing are underway, with all samples being tested in UV exposure at the time of this report. As mentioned in Section 4, the panels will be sent to Cal Poly for flexural testing. Information is still being gathered on whether there is an avenue to test tensile behavior. The prototype boat gunnel provided great steps forward in understanding the benefits and limitations of heated tooling. The biggest concern with heated tooling is maintaining better control over temperature. For more intricate parts, heating can vary substantially across different surfaces depending on factors like wire configuration. To better control the molds' temperature and avoid overheating (like Test 1), a PID controller is a much better solution than a voltage controller. Based on temperatures taken from Tests 1 and 2, at least four thermocouples would produce the best spread of temperature data points. General locations for each are highlighted below in Figure 10. It was found the corners of the mold heated up quickest, likely due to bends in the wire around those spots, and the short walls (far left circle) heated higher, too. The middle of the gunnel tended to take longer to heat, likely due to less wire density.



Figure 11. Potential thermocouple locations.

One potential PID could be the from CTDOFUM, called the [“4 Zone Channel Pid Temperature Controller SSR Relay Output 110V 220VAC 96MM 4 Channel Pid \(SSR Output\).”](#) This PID can

take 4 inputs and uses an SSR drive to control temperature, reducing mechanical relay wear. It costs about \$150 per unit, which would be good for prototype testing or smaller part creation. Larger parts may require more thermocouple connections.

Another option is the Autonics TA Series “[Analog Non-Indicating Type PID Temperature Controllers](#).” This analog system allows for control between PID and On/Off which may allow for more flexible use. The website does not openly share thermocouple connection quantity, but it could be useful for small parts where temperature is generally consistent.

From the same company, the TM Series “[Modular Multi-Channel PID Temperature Controllers](#)” may be useful for parts of the gunnel size or larger. It has a scalability factor which can allow for larger part readings. It appears to require a separate part to read temperature output, but the scalability of this unit holds good promise.

The gunnel prototype part pulled from Test 2 is not a high-quality piece and would certainly not fly as an industry product. However, it provided a lot of good data on the pros and cons of both lay-up and curing procedures.

7. Appendix

A.1 UV Panel Imaging

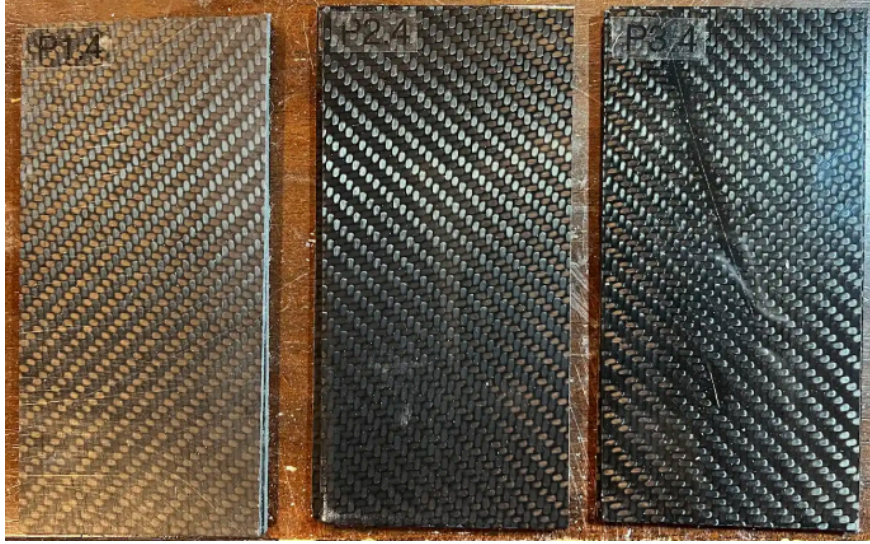


Figure A1. Prepreg panels: P1.4 - raw prepreg, P2.4 - Berg finish spray, P3.4 - PPG over Berg finish sprays.



Figure A2. Rovicore panels: R1.4 - Vinyl ester infusion, R3.4 - Proxima, R4.4 - Vinyl Ester with white gelcoat dye.

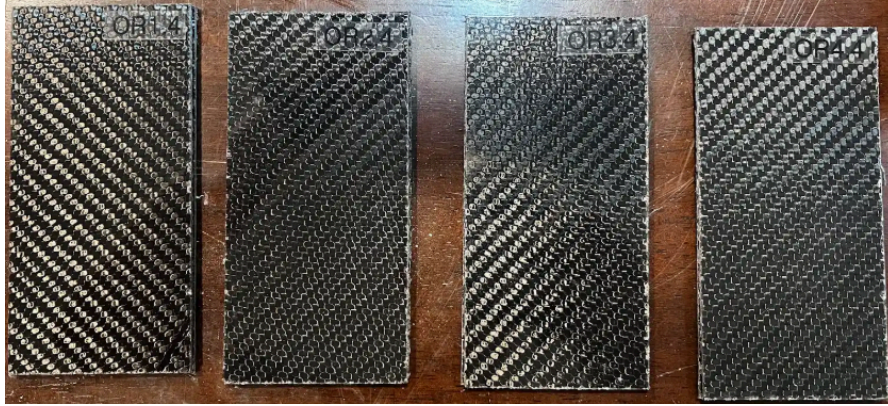


Figure A3. Redmond manufactured panels: OR1.4 - Clear polyester gelcoat with vinyl ester resin, OR2.4 - Marine gelcoat with vinyl ester resin, OR3.4 - Clear polyester gelcoat with 8086 polyester resin, OR4.4 - Marine gelcoat with 8086 polyester resin.

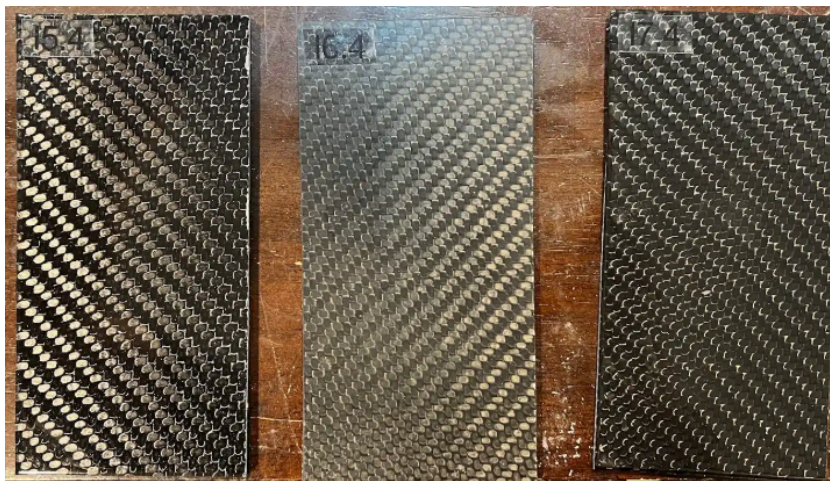
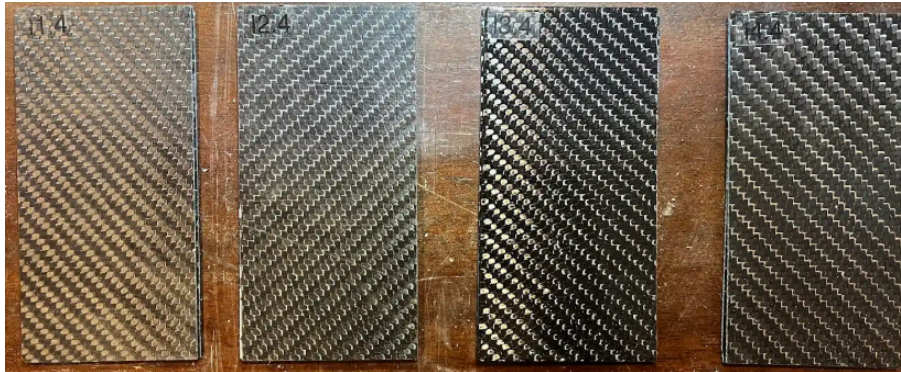


Figure A4. Dry fiber infusions: I1.4 - Raw infusion, I2.4 - Berg IMC, I3.4 - PPG finish over Berg IMC, I4.4 - PPG IMC, I5.4 - PPG finish over PPG IMC, I6.4 - Imedge gelcoat IMC, I7.4 - Sunshield finish.

A.2 Links to spreadsheets & documents

This list of spreadsheets includes any that were used in this paper, and therefore, in my internship project work.

- [Accelerated UV Testing](#) spreadsheet
- [Accelerated UV Testing document](#)
- [Boat Gunnel Testing](#)
- [Heated Mold Calculator](#)